Lecture 8

Systems of Particles

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MC2103

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Lecture 8-1

https://youtu.be/VmifDvvlawk

Lecture 8-2

Contents

Engineers often need to analyze the dynamics of systems of particles – this is the basis for many fluid dynamics applications, and will also help establish the principles used in analyzing rigid bodies.

GIST

Introduction

In the current chapter, you will study the motion of systems of particles.

- The effective force of a particle
  - The effective force of a particle is defined as the product of its mass and acceleration.
  - It will be shown that the system of external forces acting on a system of particles is equipollent with the system of $m_i \mathrm{a}_i$ for the various particles.

GIST

Introduction

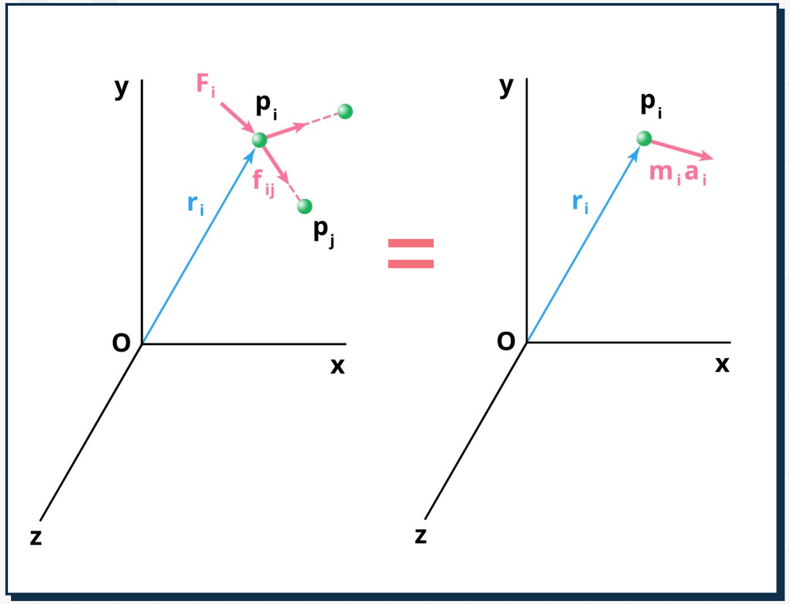
In the current chapter, you will study the motion of systems of particles.

- The mass center of a system of particles
  - The mass center of a system of particles will be defined and its motion described.

- Application of the work-energy principle and the impulse-momentum principle to a system of particles will be described.
  - Result obtained are also applicable to a system of rigidly connected particles, i.e., *a rigid body.*

GIST

Applying Newton's Law and Momentum Principles



- Newton's second law for each particle P_i in a system of n particles

$$\bar{F}_i + \sum_{j=1}^{n} \vec{f}_{ij} = m_i \vec{a}_i$$

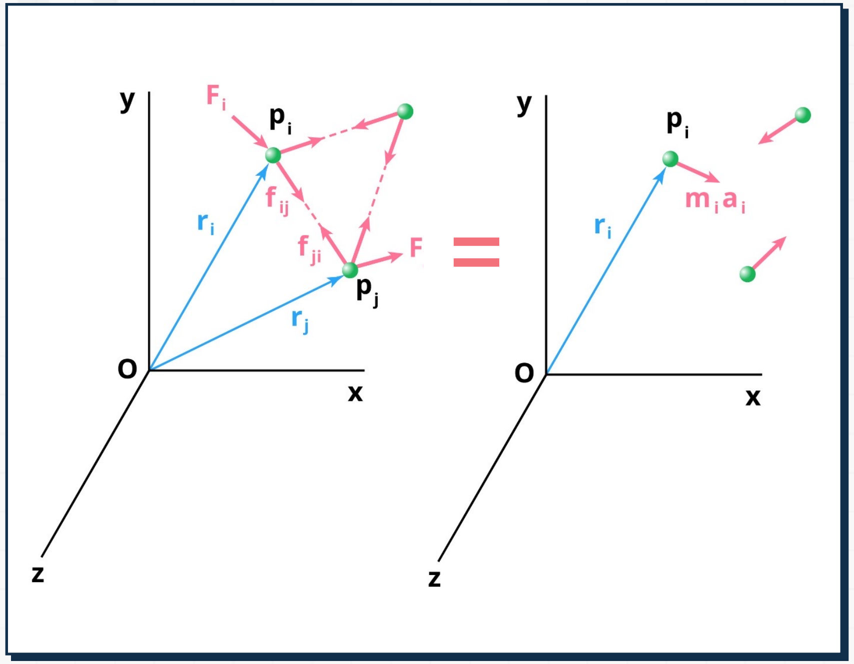
$$\bar{r}_i \times \bar{F}_i + \sum_{j=1}^{n} (\vec{r}_i \times \vec{f}_{ij}) = \vec{r}_i \times m_i \vec{a}_i$$

$\bar{F}_i$ = external force

$\bar{f}_{ij}$ = internal force

$m_i \bar{a}_i$ = external force

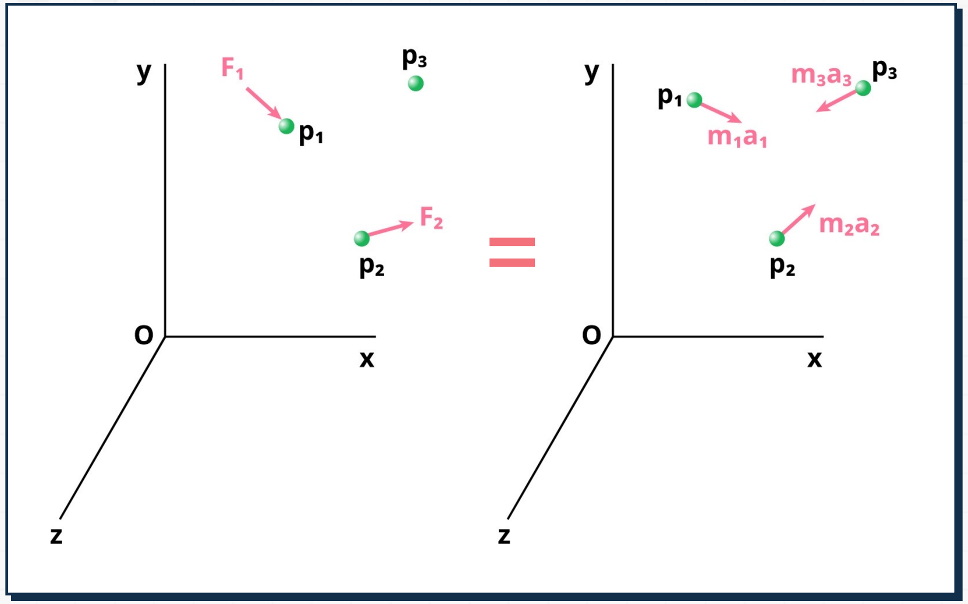
Applying Newton's Law and Momentum Principles



- The system of external and internal forces on a particle is *equivalent* to the effective force of the particle.
- The system of external and internal forces acting on the entire system of particles is *equivalent* to the system of effective forces.

GIST

Applying Newton's Law and Momentum Principles



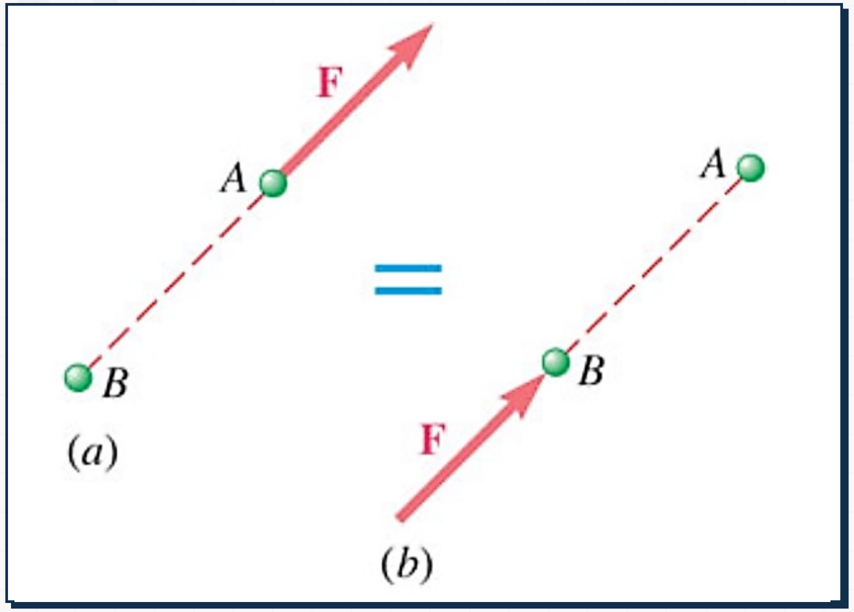
Summing over all the elements,

$$\sum_{i=1}^{n} \vec{F}_i + \sum_{i=1}^{n}\sum_{j=1}^{n} \vec{f}_{ij} = \sum_{i=1}^{n} m_i \bar{a}_i$$

$$\sum_{i=1}^{n} (\vec{r}_i \times \vec{F}_i) + \sum_{i=1}^{n}\sum_{j=1}^{n} (\vec{r}_i \times \vec{f}_{ij}) = \sum_{i=1}^{n} (\vec{r}_i \times m_i \vec{a}_i)$$

GIST

Applying Newton's Law and Momentum Principles

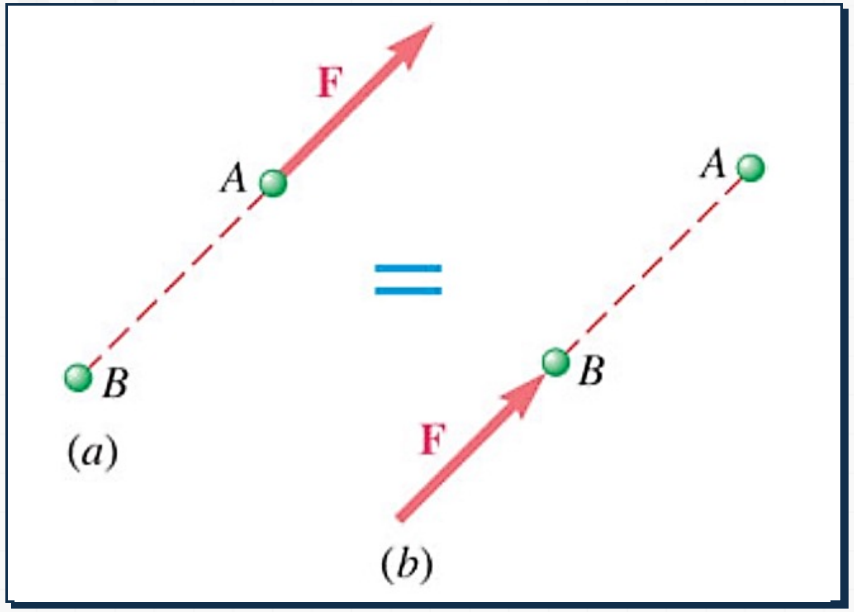


- Since the internal forces occur in equal and opposite collinear pairs, the resultant force and couple due to the internal forces are zero,

$$\sum \vec{F}_i = \sum m_i \vec{a}_i$$

$$\sum (\vec{r}_i \times \vec{F}_i) = \sum (\vec{r}_i \times m_i \vec{a}_i)$$

Applying Newton's Law and Momentum Principles



- The system of external forces and the system of $m_i \mathrm{a}_i$ are *equipollent* by not *equivalent*.

Linear & Angular Momentum

Linear Momentum

- Linear momentum of the system of particles,

$$\vec{L} = \sum_{i=1}^{n} m_i \vec{v}_i$$

$$\dot{\vec{L}} = \sum_{i=1}^{n} m_i \dot{\vec{v}}_i = \sum_{i=1}^{n} m_i \vec{a}_i$$

GIST

Linear & Angular Momentum

Linear Momentum

Resultant of the external forces

- Resultant of the external forces is equal to rate of change of linear momentum of the system of particles,

$$\sum \vec{F} = \dot{\vec{L}}$$

GIST

Linear & Angular Momentum

Angular Momentum

- Angular momentum about fixed point *O* of system of particles,

$$\vec{H}_0 = \sum_{i=1}^{n} (\vec{r}_i \times m_i \vec{v}_i)$$

$$\dot{\vec{H}}_0 = \sum_{i=1}^{n} (\dot{\vec{r}}_i \times m_i \vec{v}_i) + \sum_{i=1}^{n} (\vec{r}_i \times m_i \dot{\vec{v}}_i)$$

$$= \sum_{i=1}^{n} (\vec{r}_i \times m_i \vec{a}_i)$$

Linear & Angular Momentum

Angular Momentum

Moment resultant about fixed point O of the external forces

- Moment resultant about fixed point O of the external forces is equal to the rate of change of angular momentum of the system of particles,

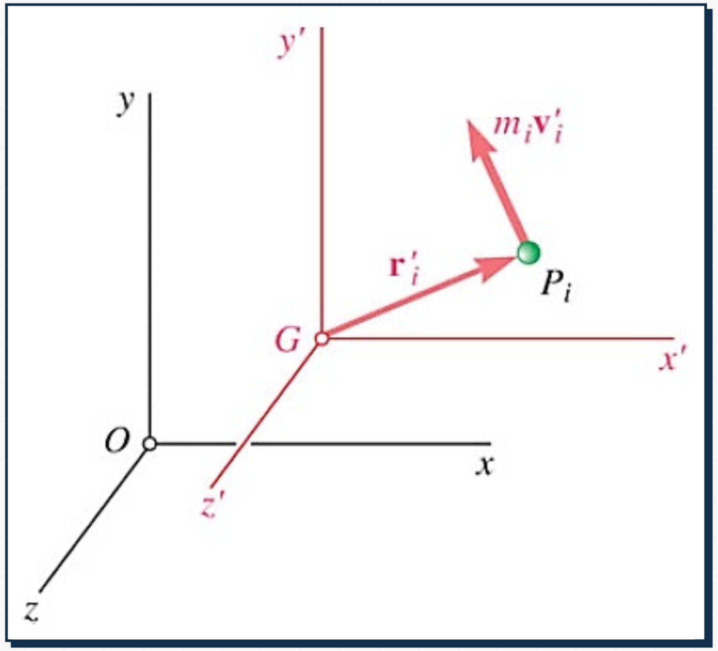
$$\sum \vec{M}_0 = \dot{\vec{H}}_0$$

Lecture 8-3

Motion of the Mass Center of a System of Particles

Mass center G of a system of particles

- Mass center G of system of particles is defined by position vector $\vec{\bar{r}}$ which satisfies.



$$m\vec{\bar{r}} = \sum_{i=1}^{n} m_i \vec{r}_i$$

Motion of the Mass Center of a System of Particles

Differentiating twice,

$$m\dot{\vec{r}} = \sum_{i=1}^{n} m_i \dot{\vec{r}}_i$$

$$m\dot{\vec{v}} = \sum_{i=1}^{n} m_i \dot{\vec{v}}_i = \dot{\vec{L}}$$

$$m\vec{a} = \dot{\vec{L}} = \sum \vec{F}$$

Motion of the Mass Center of a System of Particles

Motion of the mass center

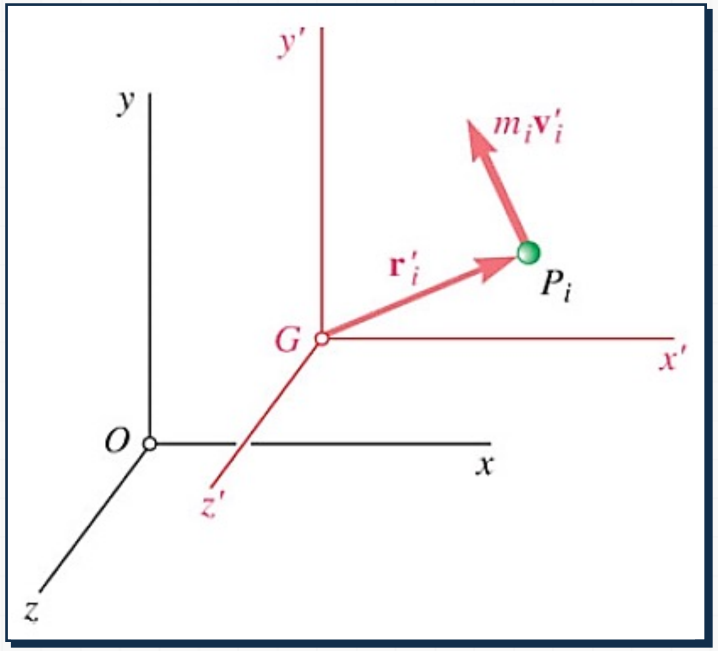
- The mass center moves as if the entire mass and all of the external forces were concentrated at that point.

GIST

Angular Momentum About the Mass Center

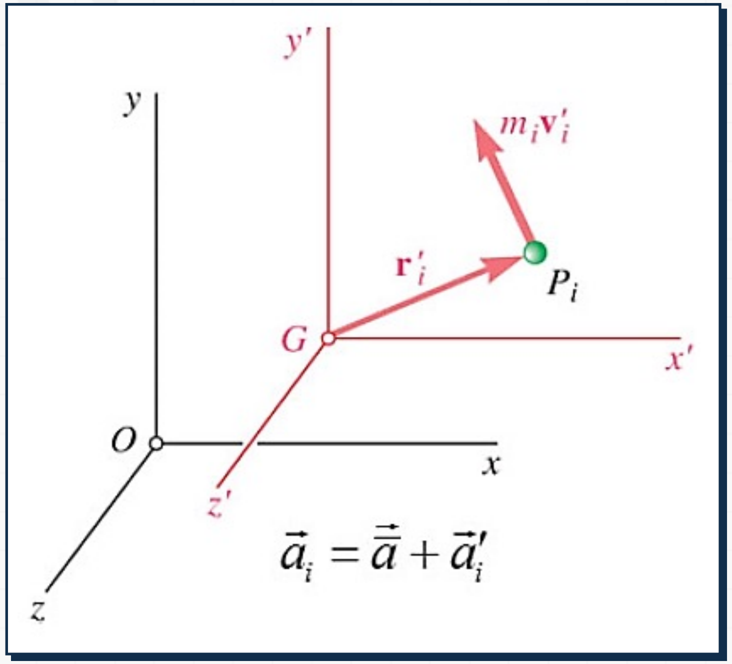
Consider the centroidal frame of reference $Gx'y'z'$, which translates with respect to the Newtonian frame $Oxyz$.

- The centroidal frame is not, in general, a Newtonian frame.



$$\vec{a}_i = \vec{\bar{a}} + \vec{a}'_i$$

Angular Momentum About the Mass Center

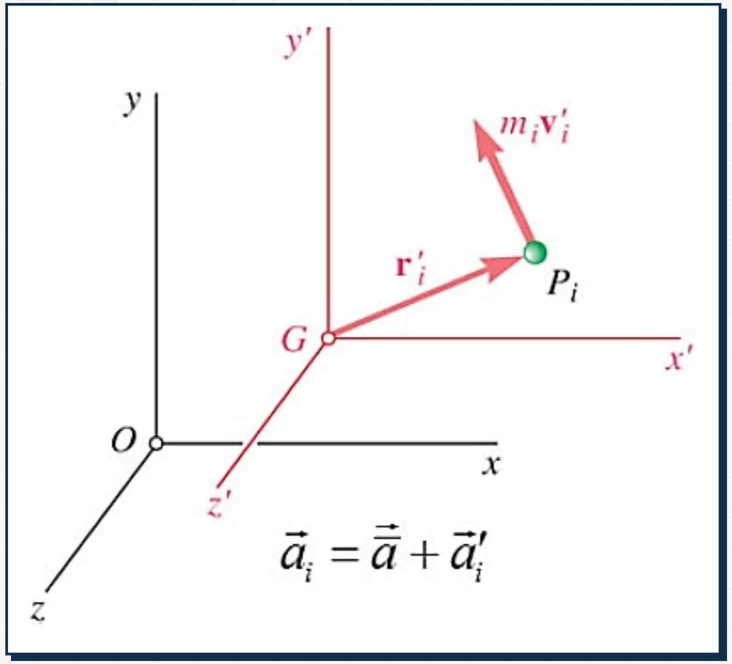


- The angular momentum of the system of particles about the mass center,

$$\vec{H}'_G = \sum_{i=1}^{n} (\vec{r}'_i \times m_i \vec{v}'_i)$$

GIST

Angular Momentum About the Mass Center

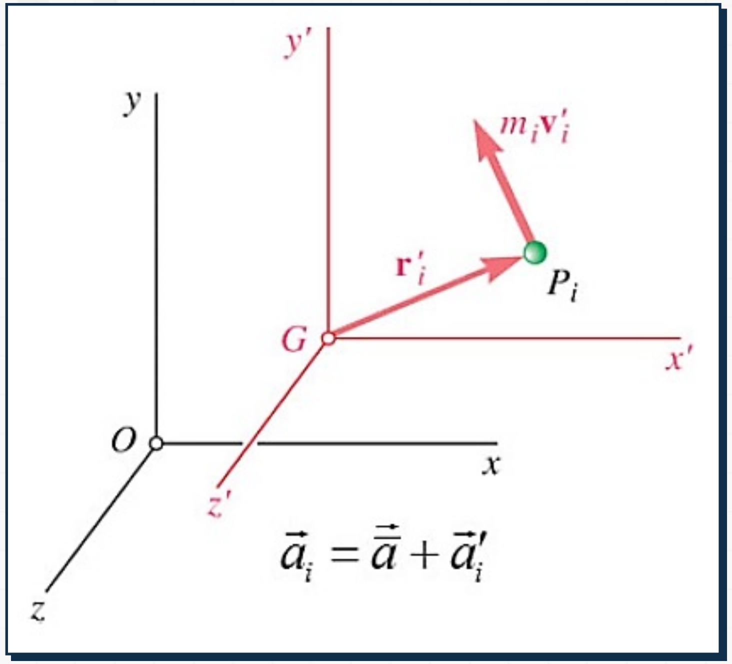


- The angular momentum of the system of particles about the mass center,

$$
\begin{aligned}
\dot{\vec{H}}'_G &= \sum_{i=1}^{n}(\vec{r}'_i \times m_i \vec{a}'_i) = \sum_{i=1}^{n}(\vec{r}_i \times m_i(\bar{a}_i - \vec{\bar{a}})) \\
&= \sum_{i=1}^{n}(\vec{r}'_i \times m_i \vec{a}_i) - \left(\sum_{i=1}^{n} m_i \vec{r}'\right) \times \vec{\bar{a}} \\
&= \sum_{i=1}^{n}(\vec{r}'_i \times m_i \vec{a}_i) = \sum_{i=1}^{n}(\vec{r}'_i \times \vec{F}_i) \\
&= \sum \vec{M}_G
\end{aligned}
$$

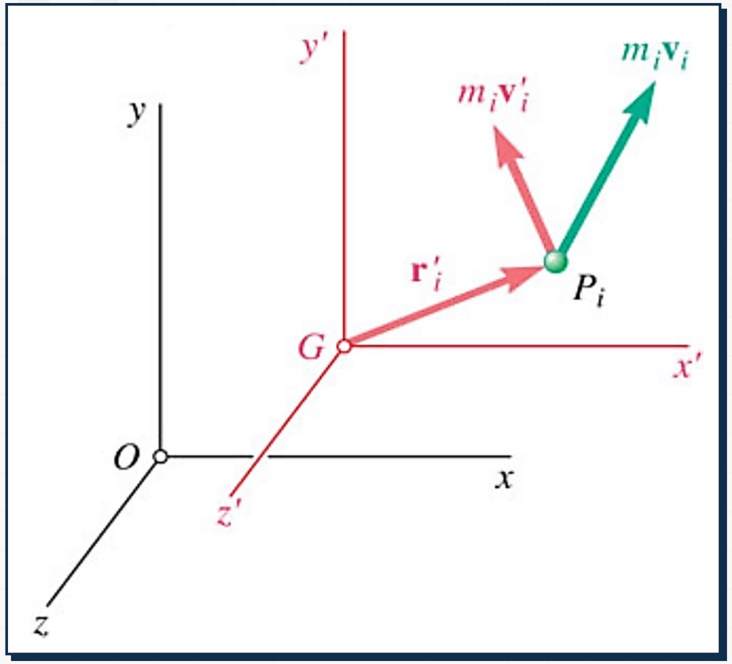
GIST

Angular Momentum About the Mass Center



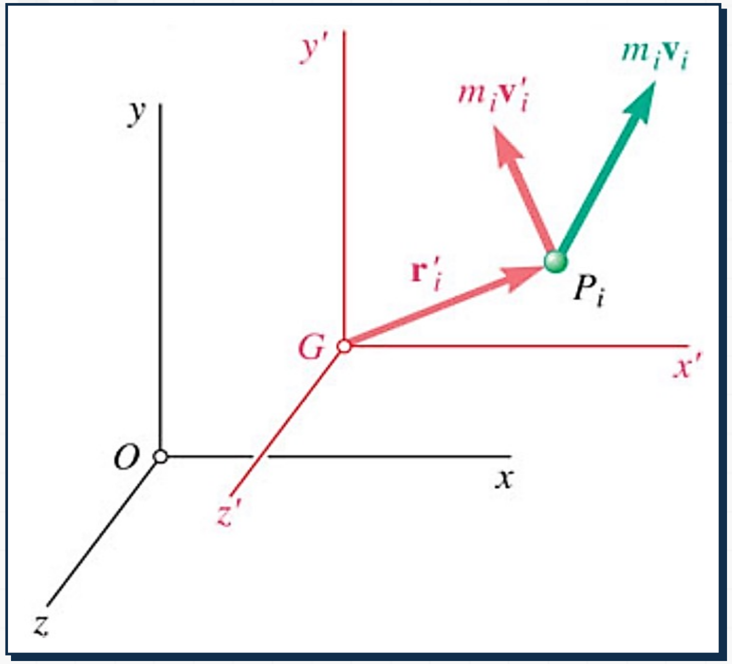
- The moment resultant about G of the external forces
  - The moment resultant about G of the external forces is equal to the rate of change of angular momentum about G of the system of particles.

Angular Momentum About the Mass Center



$$\vec{v}_i = \vec{\bar{v}} + \vec{v}'_i$$

Angular Momentum About the Mass Center

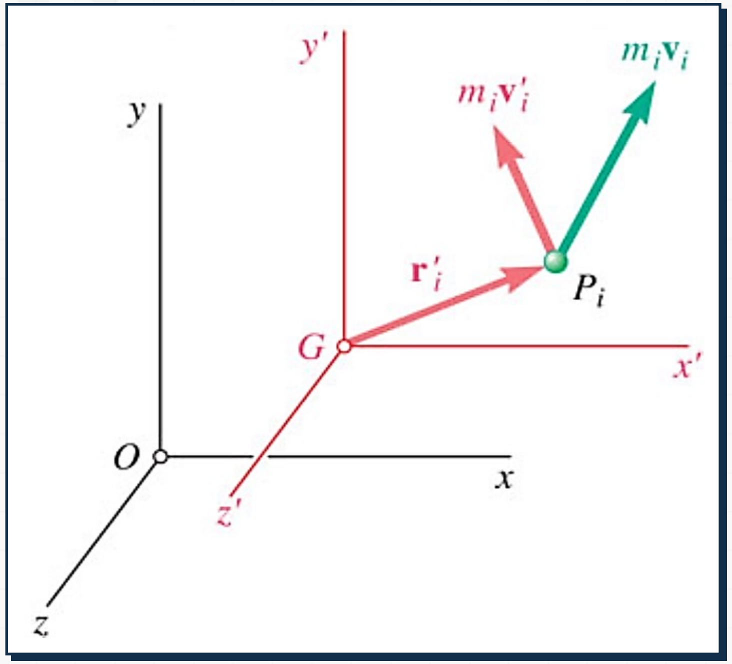


- Angular momentum about G of particles in $Gx'y'z'$
  - Angular momentum about G of the particles in their motion relative to the centroidal $Gx'y'z'$ frame of reference,

$$\vec{H}'_G = \sum_{i=1}^{n} (\vec{r}'_i \times m_i \vec{v}'_i)$$

GIST

Angular Momentum About the Mass Center



- Angular momentum about G of particles in $Oxyz$
  - Angular momentum about G of particles in their absolute motion relative to the Newtonian $Oxyz$ frame of reference.

$$\vec{H}_G = \sum_{i=1}^{n}(\vec{r}_i' \times m_i\vec{v}_i)$$

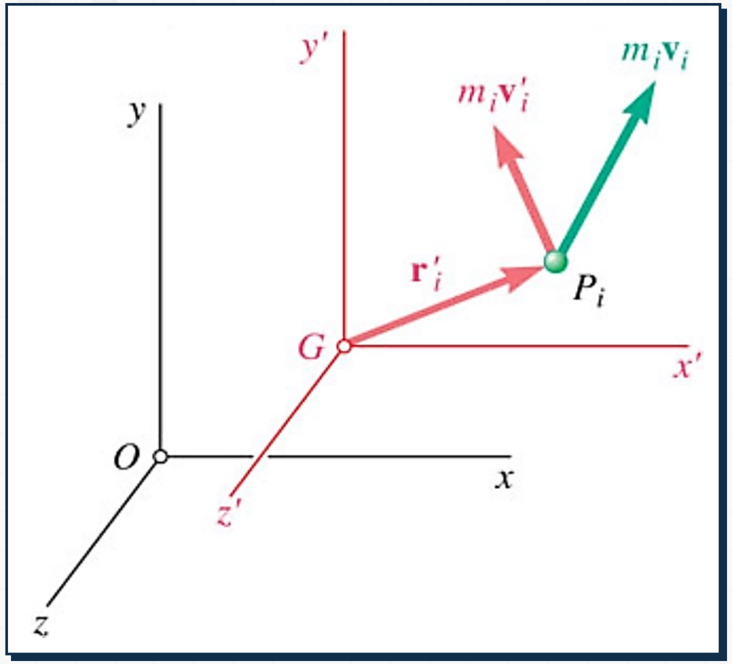
$$= \sum_{i=1}^{n}(\vec{r}_i' \times m_i(\vec{\bar{v}} + \vec{v}_i'))$$

$$= (\sum_{i=1}^{n}(m_i\vec{r}_i')\times\vec{\bar{v}} + \sum_{i=1}^{n}(\vec{r}_i' \times m_i\vec{v}_i)$$

$$\dot{\vec{H}}_G = \dot{\vec{H}}'_G = \sum \vec{M}_G$$

GIST

Angular Momentum About the Mass Center



- Angular momentum about G of the particle momenta can be calculated with respect to either the Newtonian or centroidal frames of reference.

GIST

Conservation of Momentum

If no external forces act on the particles of a system, then the linear momentum and angular momentum about the fixed point O are conserved.

$$\dot{\vec{L}} = \sum \vec{F} = 0$$

$$\vec{L} = constant$$

$$\dot{\vec{H}}_O = \sum \vec{M}_O = 0$$

$$\vec{H}_0 = constant$$

Conservation of Momentum

- **In some applications, such as problems involving central forces,**

$$\dot{\vec{L}} = \sum \vec{F} \neq 0$$

$$\vec{L} \neq constant$$

$$\dot{\vec{H}}_O = \sum \vec{M}_O = 0$$

$$\vec{H}_0 = constant$$

GIST

Conservation of Momentum

Concept of conservation of momentum

- Concept of conservation of momentum also applies to the analysis of the mass center motion,

$$\dot{\vec{L}} = \sum \vec{F} = 0$$

$$\vec{L} = m\vec{\bar{v}} = constant$$

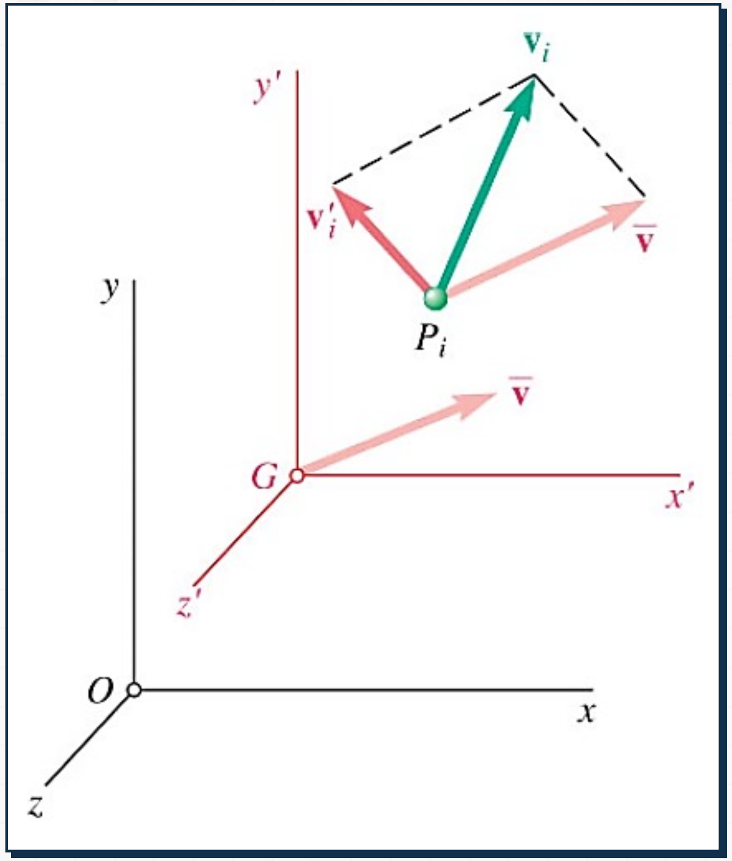
$$\vec{\bar{v}} = constant$$

$$\dot{\vec{H}}_G = \sum \vec{M}_G = 0$$

$$\vec{H}_G = constant$$

GIST

Kinetic Energy



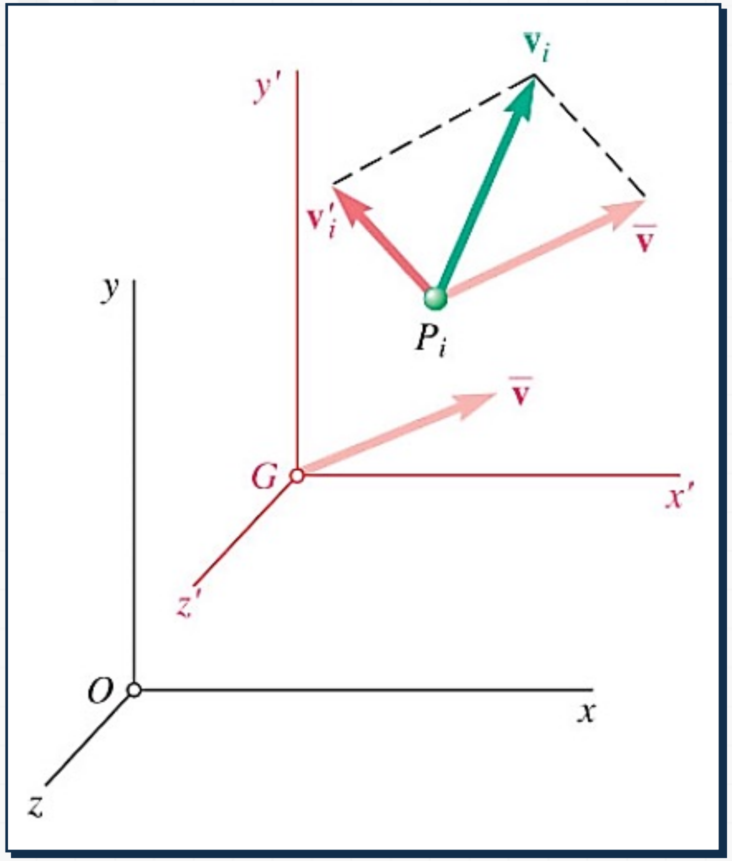
$$\vec{v}_i = \vec{v}_G + \vec{v}'_i$$

Kinetic energy of a system of particles

$$T = \frac{1}{2}\sum_{i=1}^{n} m_i\,(\vec{v}_i \bullet \vec{v}_i) = \frac{1}{2}\sum_{i=1}^{n} m_i v_i^2$$

GIST

Kinetic Energy



- Expressing the velocity in terms of the centroidal reference frame,

$$T = \frac{1}{2}\sum_{i=1}^{n}[m_i(\vec{v}_G + \vec{v}'_i) \bullet (\vec{v}_G + \vec{v}'_i)]$$

$$= \frac{1}{2}\left(\sum_{i=1}^{n} m_i\right) v_G^2 + \vec{v}_G \bullet \sum_{i=1}^{n} m_i \vec{v}'_i + \frac{1}{2}\sum_{i=1}^{n} m_i {v'_i}^2$$

$$= \frac{1}{2} m \vec{v}_G^2 + \frac{1}{2}\sum_{i=1}^{n} m_i {v'_i}^2$$

Work-Energy Principle and Conservation of Energy

Work-Energy Principle

- Principle of work and energy can be applied to each particle P_i

$$(T_1)_i + (U_{1\to 2})_i = (T_2)_i$$

where $(U_{1\to 2})_i$ represents the work done by the internal forces f_{ij} and the resultant external force F_i acting on P_i.

Work-Energy Principle and Conservation of Energy

Work-Energy Principle

- Principle of work and energy can be applied to the entire system by adding the kinetic energies of all particles and considering the work done by all external and internal forces.
- Although f_{ij} and f_{ji} are equal and opposite, the work of these forces will not, in general, cancel out since the displacement for P_i and P_j are different.

Work-Energy Principle and Conservation of Energy

Conservation of Energy

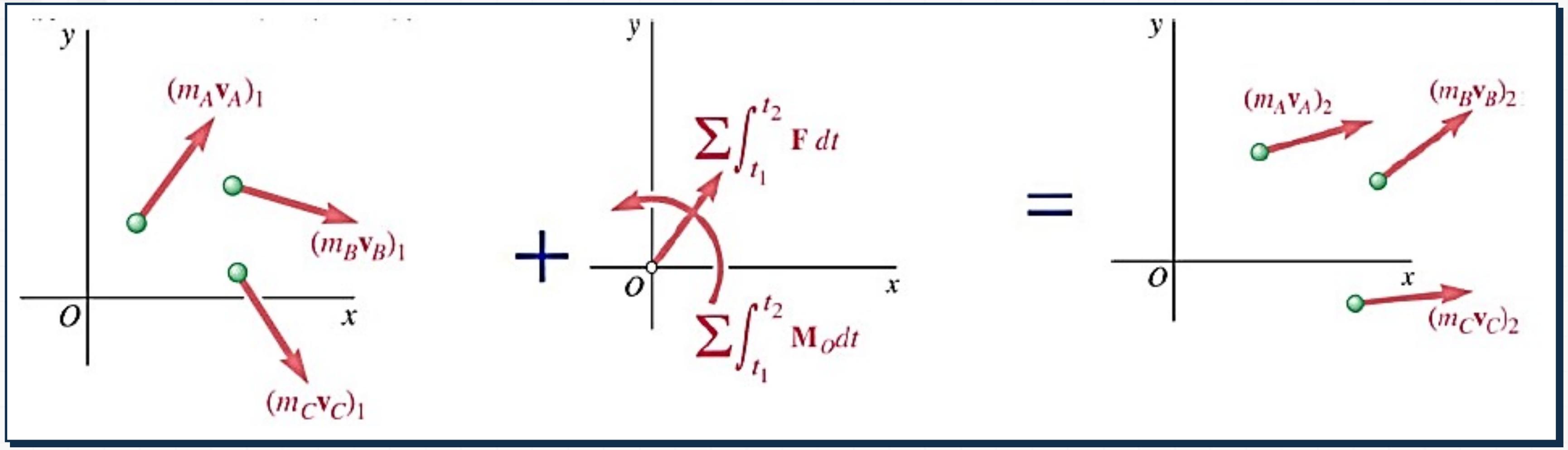
- If the forces acting on the particles are conservative, the work is equal to the change in potential energy and

$$T_1 + V_1 = T_2 + V_2$$

which expresses the principle of conservation of energy for the system of particles.

GIST

Impulse-Momentum Principle



$$\sum \vec{F} = \dot{\vec{L}}$$

$$\sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2 - \vec{L}_1$$

$$\vec{L}_1 + \sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2$$

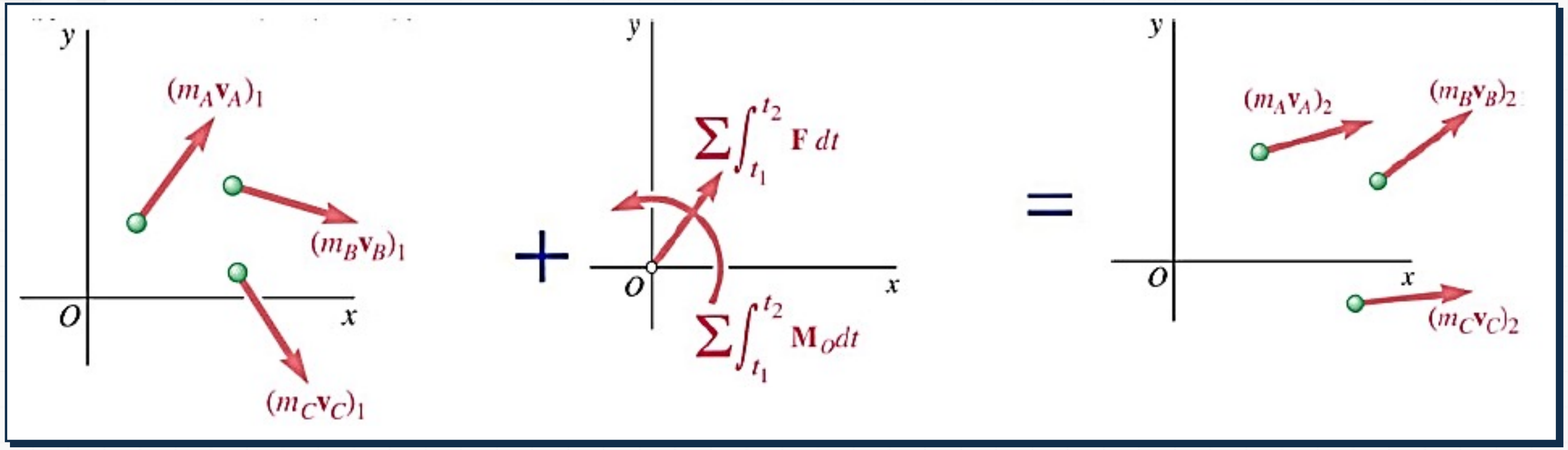
$$\sum \vec{M}_O = \dot{\vec{H}}_O$$

$$\sum \int_{t_1}^{t_2} \vec{M}_O dt = \vec{H}_2 - \vec{H}_1$$

$$\vec{H}_1 + \sum \int_{t_1}^{t_2} \vec{M}_O dt = \vec{H}_2$$

GIST

Impulse-Momentum Principle



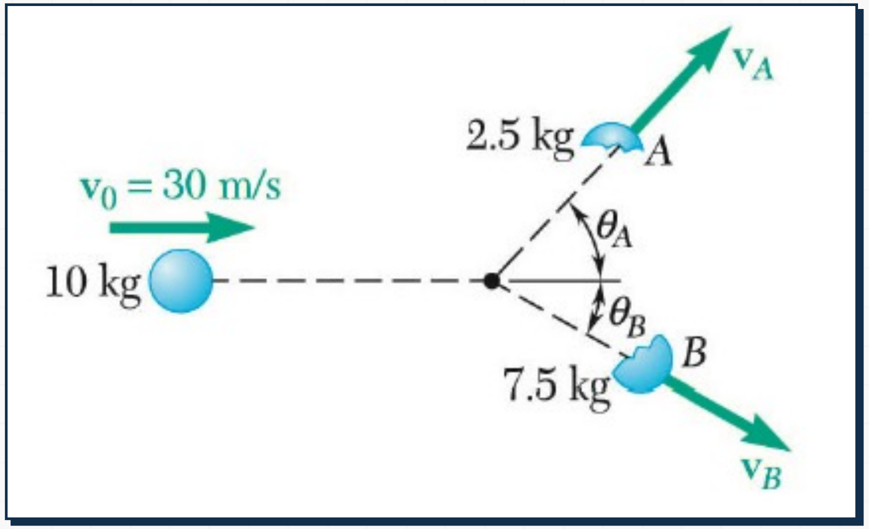
- The momenta of the particles at time t_1 and the impulse of the forces from t_1 to t_2 form a system of vectors *equipollent* to the system of momenta of the particles at time t_2.

Lecture 8-4

GIST

Sample Problem 1

- **A 10-kg projectile is moving with a velocity of 30 m/s when it explodes into 2.5 and 7.5-kg fragments.**



- Determine
  - Immediately after the explosion, the fragments travel in the directions $\theta_A = 45°$ and $\theta_B = 30°$.
  - Determine the velocity of each fragment.

GIST

Sample Problem 1

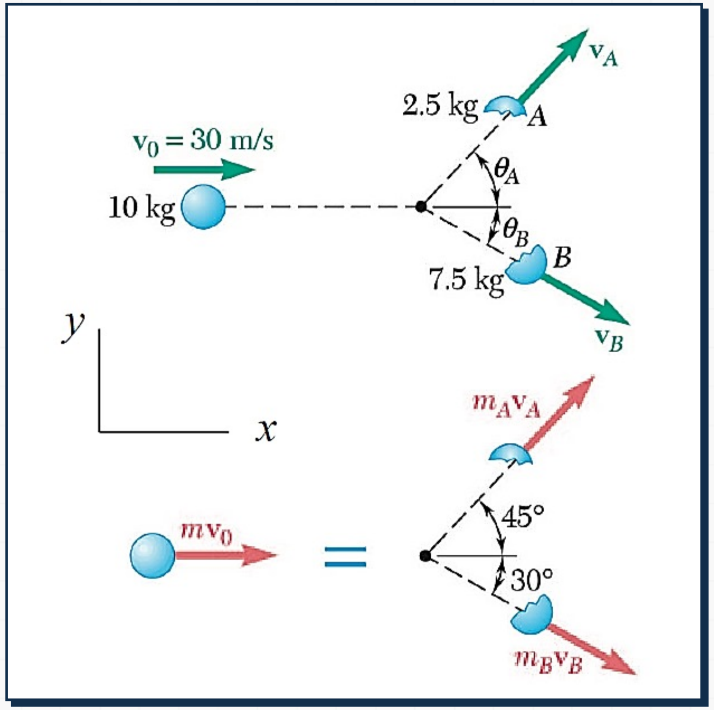
- **A 10-kg projectile is moving with a velocity of 30 m/s when it explodes into 2.5 and 7.5-kg fragments.**
  - Strategy
    - Since there are no external forces, the linear momentum of the system is conserved.
    - Write separate component equations for the conservation of linear momentum.
    - Solve the equations simultaneously for the fragment velocities.

GIST

Sample Problem 1

Modeling and Analysis

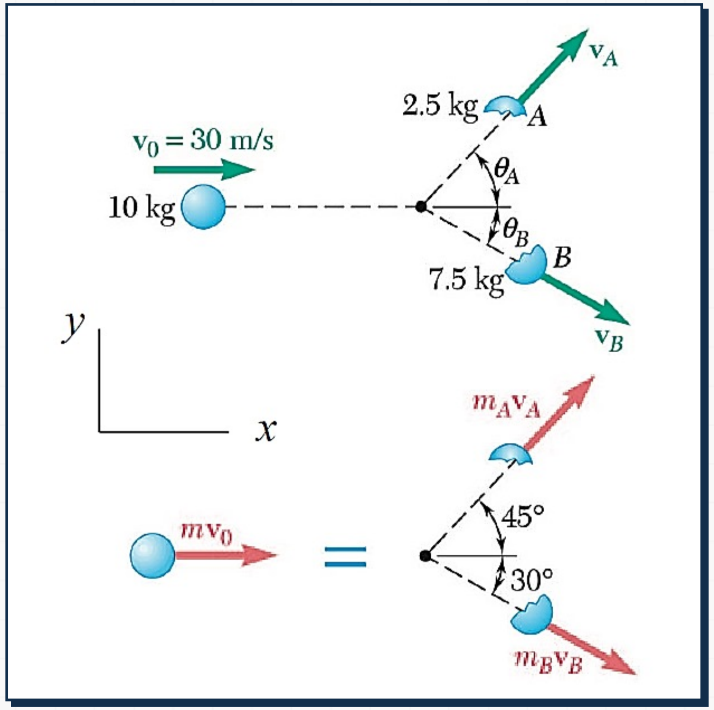
- Since there are no external forces, the linear momentum of the system is conserved.



GIST

Sample Problem 1

Modeling and Analysis



- Write separate component equations for the conservation of linear momentum.

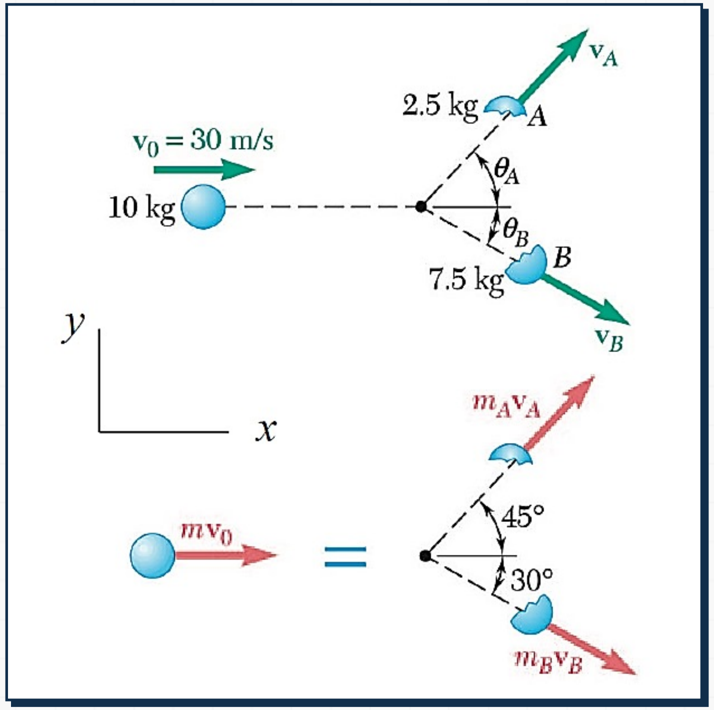
$$m_A\vec{v}_A + m_B\vec{v}_B = m\vec{v}_0$$

$$(2.5)\vec{v}_A + (7.5)\vec{v}_B = (10)\vec{v}_0$$

GIST

Sample Problem 1

Modeling and Analysis



- Write separate component equations for the conservation of linear momentum.

x components

$$2.5v_A \cos 45° + 7.5v_B \cos 30° = 10\times30$$

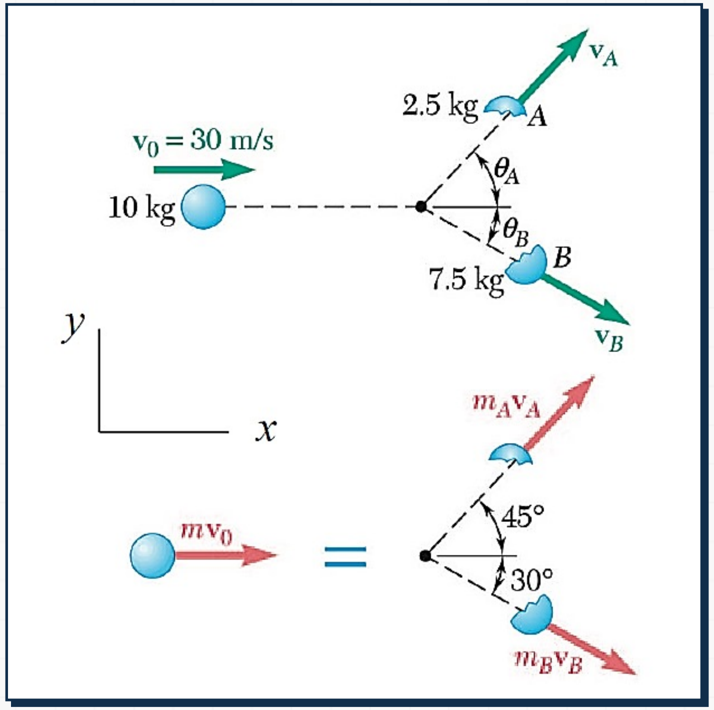
y components

$$2.5v_A \sin 45° + 7.5v_B \sin 30° = 0$$

GIST

Sample Problem 1

Modeling and Analysis



- Solve the equations simultaneously for the fragment velocities.

$$V_A = 62.2m/s \qquad V_B = 29.3m/s$$

GIST

Sample Problem 1

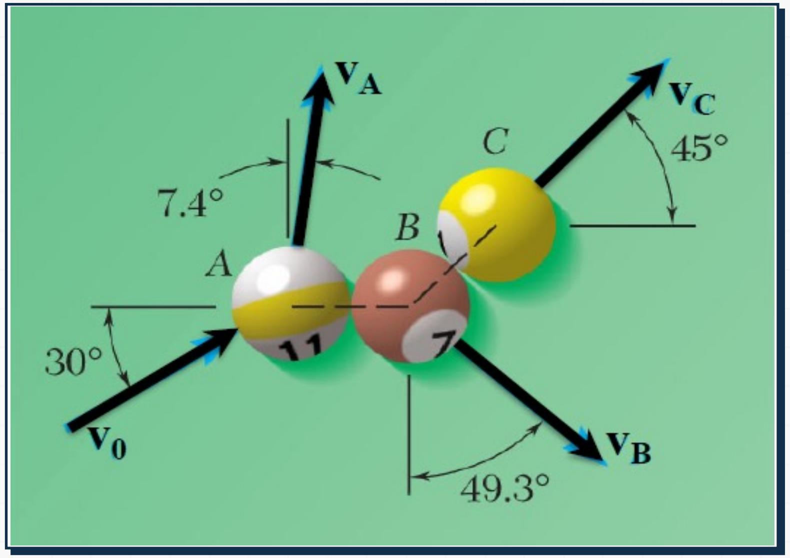
Reflect and Think

As you might have predicted, the less massive fragment winds up with a larger magnitude of velocity and departs the original trajectory at a larger angle.

GIST

Sample Problem 2

- **In a game of pool, ball A is moving with a velocity $\mathbf{v}_0$ when it strikes balls B and C, which are at rest and aligned as shown.**



- Determine
  - Knowing that after the collision the three balls move in the directions indicated and that $v_0 = 4m/s$ and $v_C = 2m/s$, determine the magnitude of the velocity of
    (a) ball A
    (b) ball B

Sample Problem 2

- **In a game of pool, ball A is moving with a velocity $\mathbf{v}_0$ when it strikes balls B and C, which are at rest and aligned as shown.**
  - Strategy
    - Since there are no external forces, the linear momentum of the system is conserved.
    - Write separate component equations for the conservation of linear momentum.
    - Solve the equations simultaneously for the pool ball velocities.

Sample Problem 2

Modeling And Analysis

- Write separate component equations for the conservation of linear momentum

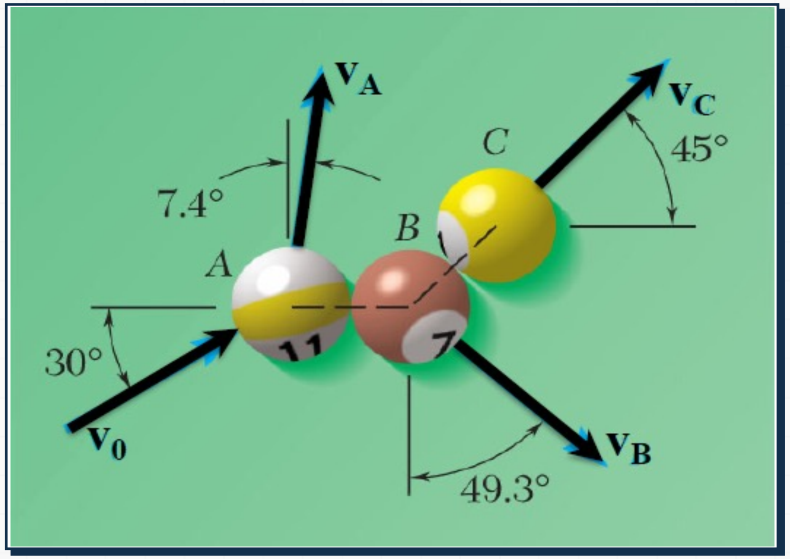
x	$m(4)\cos 30° = mv_A \sin 7.4° + mv_B \sin 49.3° + m(2)\cos 45°$ $0.12880v_A + 0.75813v_B = 2.0499 \qquad (1)$

y	$m(4)\sin 30° = mv_A \cos 7.4° + mv_B \cos 49.3° + m(2)\sin 45°$ $0.99167v_A + 0.65210v_B = 0.5858 \qquad (2)$

GIST

Sample Problem 2

Modeling And Analysis



- Two equations, two unknowns - solve

$$0.65210(0.12880V_A + 0.75813V_B = 2.0499)$$
$$+\ 0.75813(0.99167V_A + 0.65210V_B = 0.5858)$$
$$0.83581v_A = 1.78085$$

- Sub into (1) or (2) to get v_B

$$V_A = 2.13\ m/s$$
$$V_B = 2.34\ m/s$$

GIST

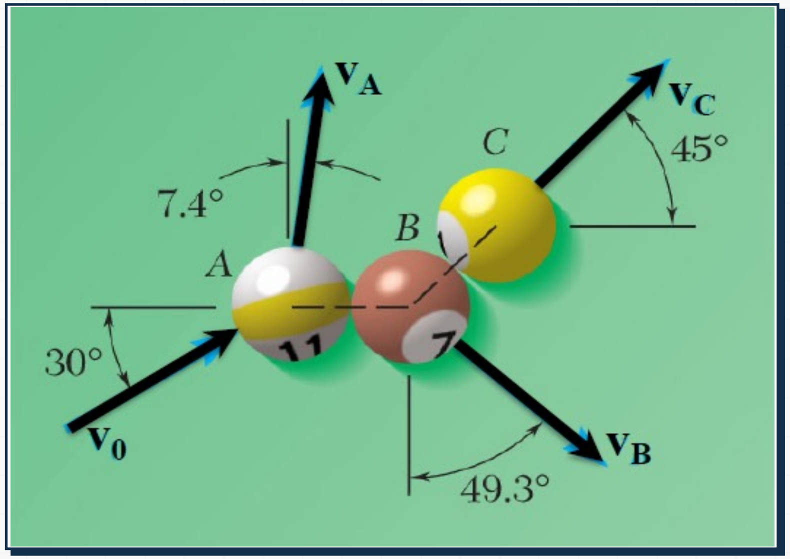
Sample Problem 2

Reflect and Think

In a game of pool, ball A is moving with a velocity v_0 when it strikes balls B and C, which are at rest and aligned as shown.

Sample Problem 2

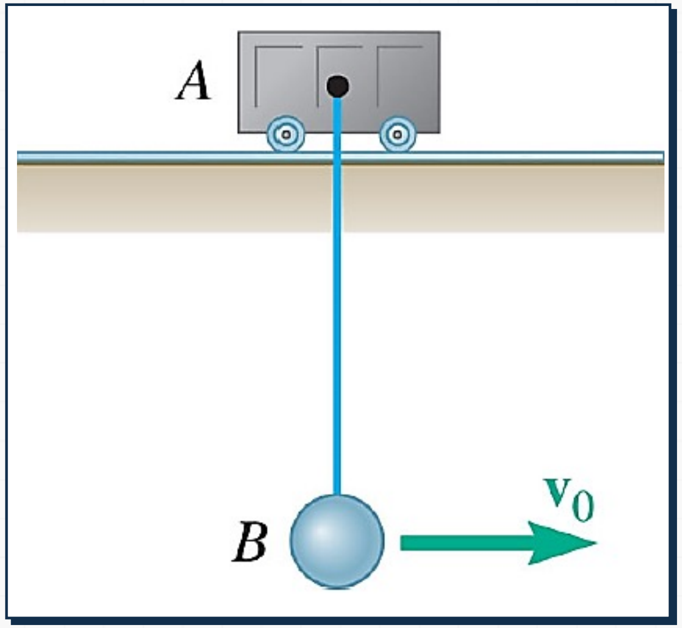
Reflect and Think



- After the impact, what is true about the overall center of mass of the system of three balls?
  - a) The overall system CG will move in the same direction as v_0
  - b) The overall system CG will stay at a single, constant point
  - c) There is not enough information to determine the CG location

Sample Problem 3

- **Ball B, of mass m_B, is suspended from a cord, of length l, attached to cart A, of mass m_A, which can roll freely on a frictionless horizontal tract.**



- Determine
  - While the cart is at rest, the ball is given an initial velocity $v_0 = \sqrt{2gl}$.
    (a) the velocity of B as it reaches it maximum elevation.
    (b) the maximum vertical distance h through which B will rise.

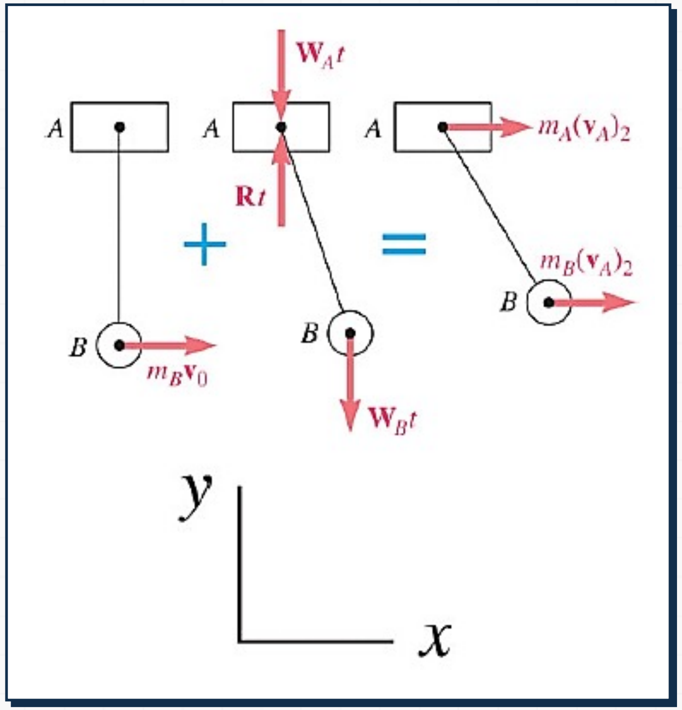
GIST

Sample Problem 3

- **Ball B, of mass m_B, is suspended from a cord, of length l, attached to cart A, of mass m_A, which can roll freely on a frictionless horizontal tract.**
  - Strategy
    - With no external horizontal forces, it follows from the impulse and momentum principle that the horizontal component of momentum is conserved. This relation can be solved for the velocity of B at its maximum elevation.
    - The conservation of energy principle can be applied to relate the initial kinetic energy to the maximum potential energy. The maximum vertical distance is determined from this relation.

Sample Problem 3

Modeling and Analysis



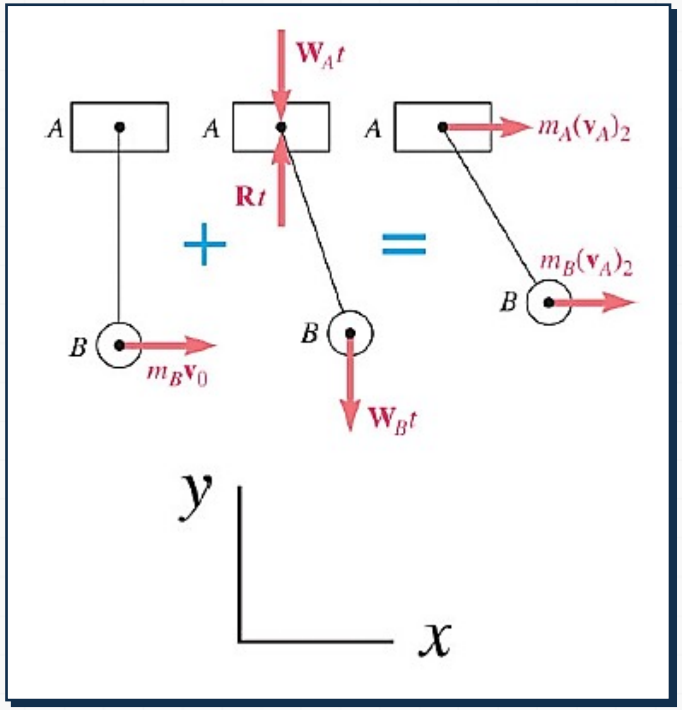
- With no external horizontal forces, it follows from the impulse-momentum principle that the horizontal component of momentum is conserved.
- This relation can be solved for the velocity of B at its maximum elevation.

$$L_1 + \int_{t_1}^{t_2} F dt = L_2$$

GIST

Sample Problem 3

Modeling and Analysis

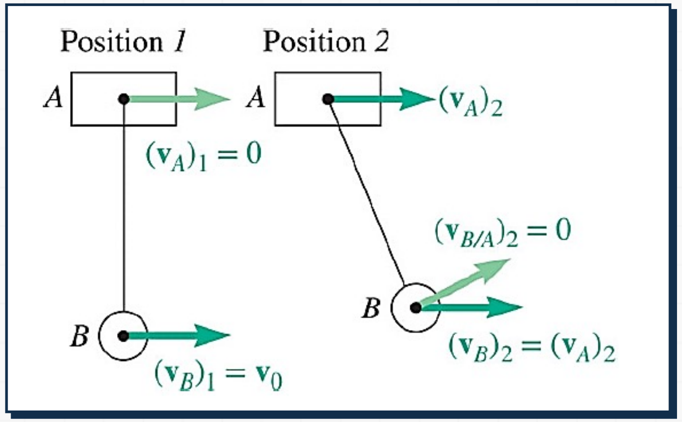


x component equation

$$m_A v_{A,1} + m_B v_{B,1} = m_A v_{A,2} + m_B v_{B,2}$$

Sample Problem 3

Modeling and Analysis



- Velocities at positions 1 and 2 are

$$v_{A,1} = 0 \qquad v_{B,1} = v_0$$

$$v_{B,2} = v_{A,2} + v_{B/A,2} = v_{A,2}$$

(velocity of B relative to A is zero at position 2)

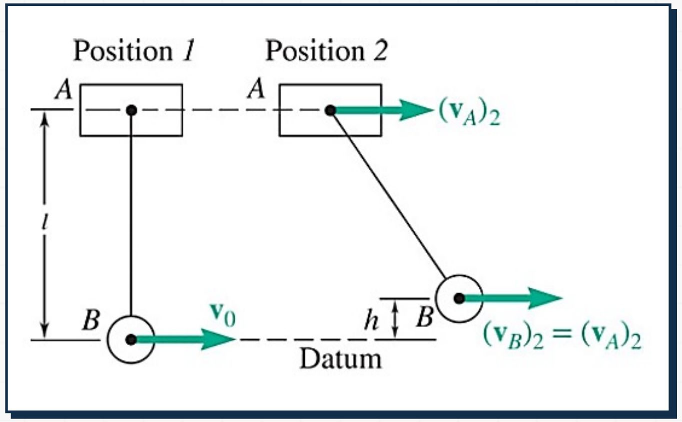
$$m_B v_0 = (m_A + m_B) v_{A,2}$$

$$v_{A,2} = v_{B,2} = \frac{m_B}{m_A + m_B} v_0$$

GIST

Sample Problem 3

Modeling and Analysis



- The conservation of energy principle can be applied to relate the initial kinetic energy to the maximum potential energy.

$$T_1 + V_1 = T_2 + V_2$$

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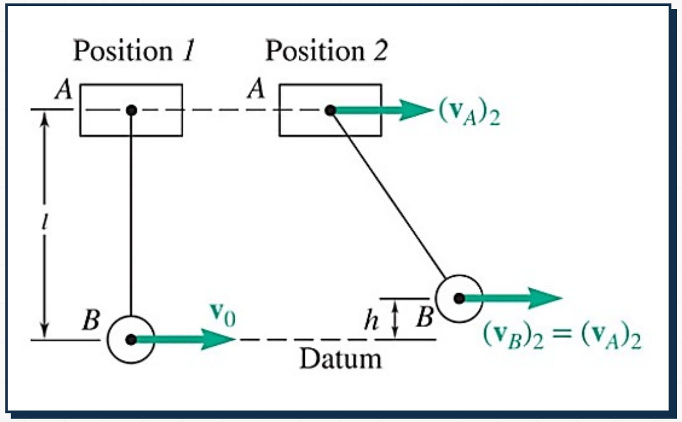
Sample Problem 3

Modeling and Analysis



- Position 1

Potential Energy	Kinetic Energy
$V_1 = m_A g l$	$T_1 = \frac{1}{2} m_B v_0^2$

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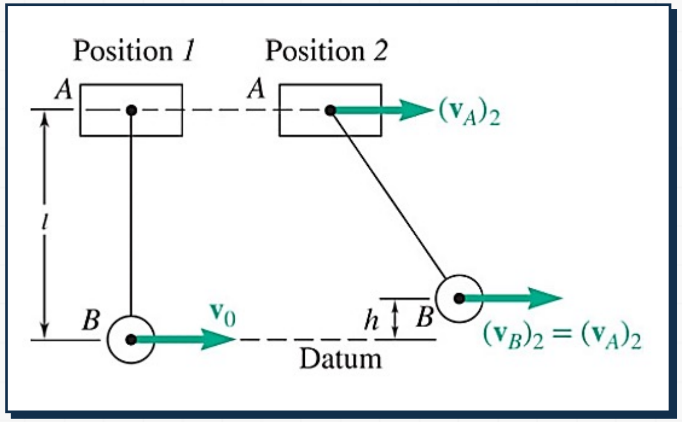
Sample Problem 3

Modeling and Analysis



- Position 2

Potential Energy	Kinetic Energy
$V_2 = m_A g l + m_B g h$	$T_2 = \frac{1}{2}(m_A + m_B)v_{A,2}^2$

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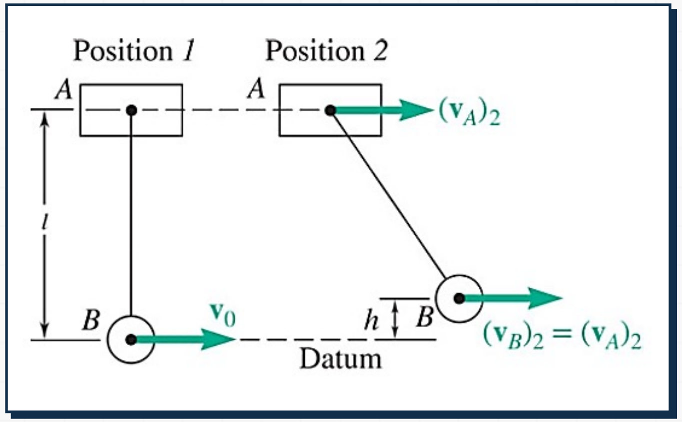
Sample Problem 3

Modeling and Analysis



$$\frac{1}{2}m_B v_0^2 + m_A g l = \frac{1}{2}(m_A + m_B)v_{A,2}^2 + m_A g l + m_B g h$$

$$h = \frac{v_0^2}{2g} - \frac{m_A + m_B}{m_B}\frac{v_{A,2}^2}{2g} = \frac{v_0^2}{2g} - \frac{m_A + m_B}{2g m_B}\left(\frac{m_B}{m_A + m_B}v_0\right)^2$$

$$h = \frac{v_0^2}{2g} - \frac{m_B}{m_A + m_B}\frac{v_0^2}{2g}$$

$$\boxed{h = \frac{m_A}{m_A + m_B}\frac{v_0^2}{2g}}$$

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Sample Problem 3

Reflect and Think

Recalling that $v_0^2 < 2gl$, it follows from the last equation that $h < 1$; this verifies that B stays below A, as assumed in the solution.

For $m_A \gg m_B$, the answers reduce to $(v_B)_2 = (v_A)_2 = 0$ and $h = v_0^2/2g$; B oscillates as a simple pendulum with A fixed.

For $m_A \ll m_B$, they reduce to $(v_B)_2 = (v_A)_2 = v_0$ and $h = 0g$; A and B move with the same constant velocity v_0.

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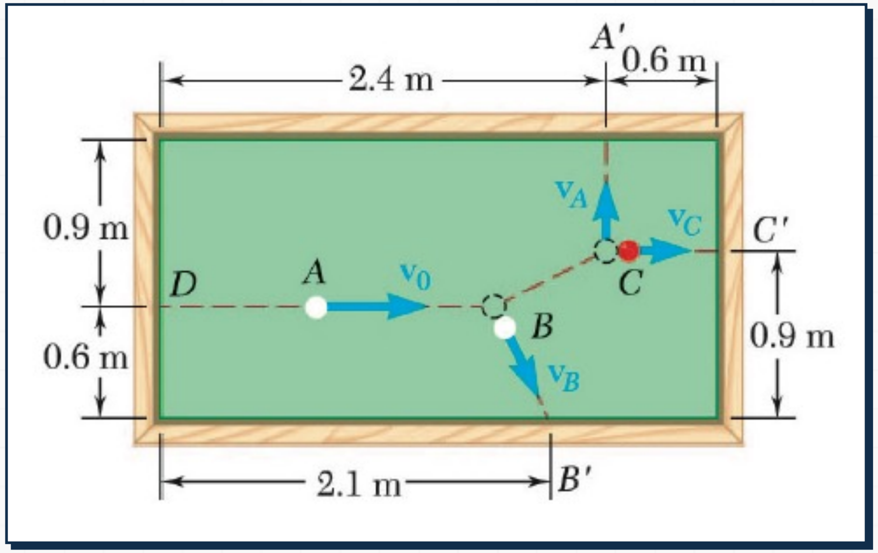
Sample Problem 4

- **Ball A has initial velocity $v_0 = 3m/s$ parallel to the axis of the table.**



- Determine
  - It hits ball B and then ball C which are both at rest.
  - Balls A and C hit the sides of the table squarely at A' and C' and ball B hits obliquely at B'.
  - Assuming perfectly elastic collisions, determine velocities v_A, v_B, and v_C with which the balls hit the sides of the table.

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Sample Problem 4

- **Ball A has initial velocity $v_0 = 3m/s$ parallel to the axis of the table.**
  - Strategy
    - There are four unknowns: $v_A, v_{B,x}, v_{B,y}$, and v_C.
    - Solution requires four equations: conservation principles for linear momentum (two component equations), angular momentum, and energy.
    - Write the conservation equations in terms of the unknown velocities and solve simultaneously.

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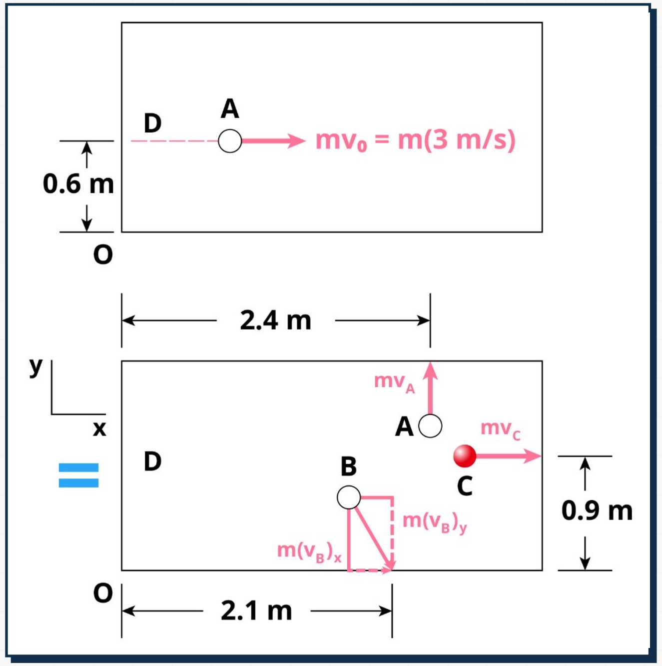
Sample Problem 4

Modeling and Analysis

- There are four unknowns: v_A, $v_{B,x}$, $v_{B,y}$, and v_C.



$$\vec{v}_A = v_A \vec{j}$$

$$\vec{v}_B = v_{B,x}\vec{i} + v_{B,y}\vec{j}$$

$$\vec{v}_C = v_{C}\vec{i}$$

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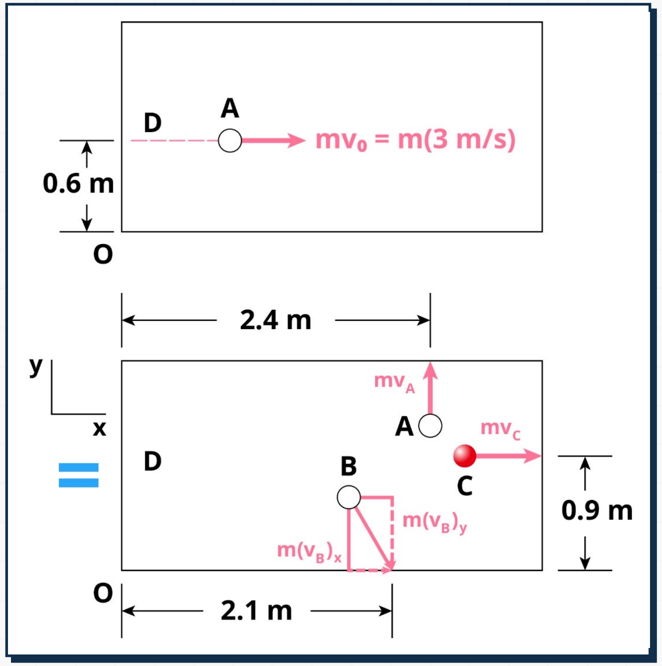
Sample Problem 4

Modeling and Analysis



- The conservation of momentum and energy equations,

$$\vec{L}_1 = \sum \int \vec{F}dt = \vec{L}_2$$

$$mv_0 = mv_{B,x} + mv_C \qquad 0 = mv_A - mv_{B,y}$$

$$\vec{H}_{O,1} + \int \vec{M}_O dt = \vec{H}_{O,2}$$

$$-(0.6)mv_0 = (2.4)mv_A - (2.1)mv_{B,y} - (0.9)mv_C$$

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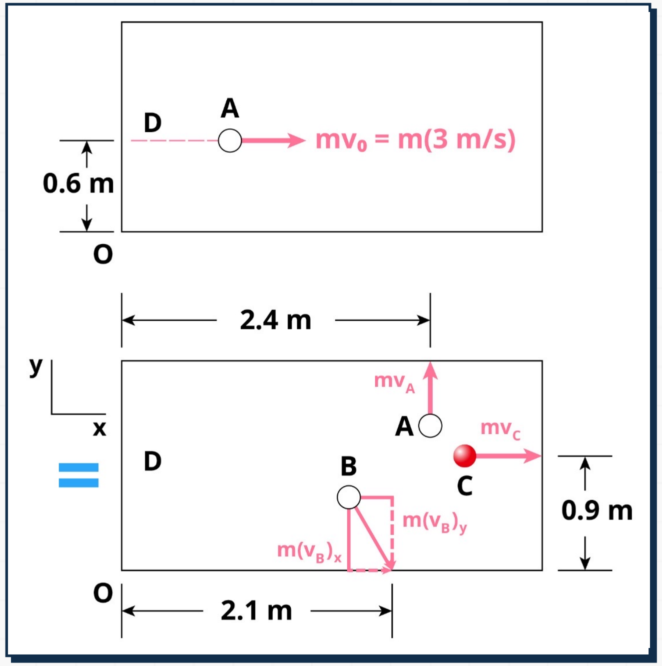
Sample Problem 4

Modeling and Analysis



- The conservation of energy equation due to perfectly elastic collision,

$$T_1 + V_1 = T_2 + V_2$$

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}m\left(v_{B,x}^2 + v_{B,y}^2\right) + \frac{1}{2}mv_C^2$$

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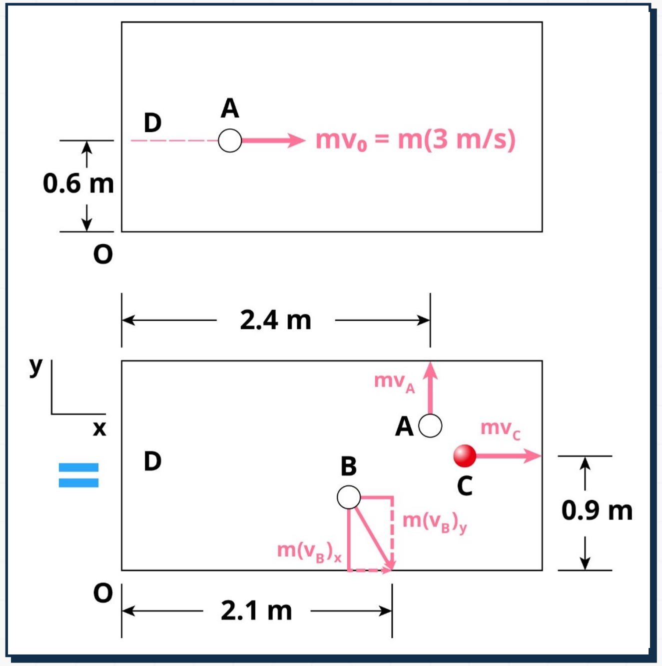
Sample Problem 4

Modeling and Analysis



- Solving the first three equations in terms of v_C,

$$V_A = V_{B,y} = 3v_C - 6 \qquad V_{B,x} = 3 - v_C$$

Substituting into the energy equation,

$$2(3v_C - 6)^2 + (3 - v_C)^2 + v_C^2 = 9$$

$$20v_C^2 - 78v_C + 72 = 0$$

$$v_A = 1.2 m/s \qquad v_C = 2.4 m/s$$

$$\vec{v}_B = (0.6\hat{\imath} - 1.2\hat{\jmath}) m/s \qquad v_B = 1.342\ m/s$$

Sample Problem 4

Reflect and Think

In a real situation, energy would not be conserved, and you would need to know the coefficient of restitution between the balls to solve this problem.

We also neglected friction and the rotation of the balls in our analysis, which is often a poor assumption in pool or billiards. We discuss rigid-body impacts in Chapter 17.

GIST

References

- Beer, Johnston, Cornwell, Self, Sanghi, “Vector Mechanics for Engineers: Dynamics” 12th Edition, Chapter 14